Breastfeeding and Cognitive Development

PROBIT RCT and sibling-comparison evidence indicate a small, verbal-concentrated, fading causal effect of breastfeeding on IQ.

GRADE: MODERATE | STATUS: DRAFT | DOMAIN: FEEDING, COGNITION | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Observational studies consistently show a 3-4 IQ-point advantage for breastfed children, but the strongest causal designs (PROBIT cluster-randomized trial follow-up, sibling comparisons) indicate the long-term causal effect for full-term infants is small, concentrated in verbal ability, and may approach zero by adolescence.

EVIDENCE AND CAUSAL ASSESSMENT

- PROBIT (Belarus, N=17,046 randomized at the hospital/cluster level) found infants in the breastfeeding-promotion arm scored 5.9 points higher full-scale IQ (not significant) and 7.5 points higher verbal IQ (significant) at age 6.5.

- 16-year PROBIT follow-up (79.5% retention, N=13,557) found no significant difference in overall neurocognitive function; only verbal function remained significantly higher (+1.4 points).

- Sibling-comparison studies (Der 2006; Colen & Ramey 2014) using within-family designs to remove shared confounders found no significant cognitive differences between breastfed and non-breastfed siblings.

EFFECT SIZE AND PERSISTENCE

Randomized estimate at age 6.5: +5.9 full-scale IQ (ns), +7.5 verbal IQ (significant). At age 16: overall effect non-significant; verbal effect fades to +1.4 points. Sibling designs: null. Best summary estimate for full-term, adequately nourished infants: 0-2 IQ points, likely concentrated in verbal domain, with fade-out over adolescence.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: largely confounded (>50%) (observational IQ gains largely attenuate in sibling-comparison designs and PROBIT trials).

Partially-to-largely confounded relative to observational estimates: PROBIT's cluster-randomized design removes maternal-choice confounding, yet the effect that remains after 16 years is far smaller than initial observational estimates (~3-4 points) and disappears in sibling comparisons. Findings may not generalize to preterm or malnourished populations.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not support claims that breastfeeding produces large or lifelong IQ gains, nor that formula-feeding meaningfully harms cognitive development in full-term, well-nourished populations.

WHAT ACTION IS REASONABLY JUSTIFIED

Breastfeeding promotion is justifiable on other well-established grounds (infection reduction, maternal health, cost), not primarily on cognitive-enhancement grounds for full-term infants in high-resource settings.

VERIFIED SOURCES & DOI IDENTIFIERS

- Kramer et al. (2008), Arch Gen Psychiatry, PROBIT IQ at 6.5 years
- Yang et al. (2018), PLOS Medicine, 16-year PROBIT follow-up
- Der, Batty & Deary (2006), BMJ, sibling pairs and meta-analysis
- Colen & Ramey (2014), Soc Sci Med, sibling comparisons
- Evenhouse & Reilly (2005), sibling comparisons (catalog ID sibling-comparison-2006)